UNDERSTANDING CELLS

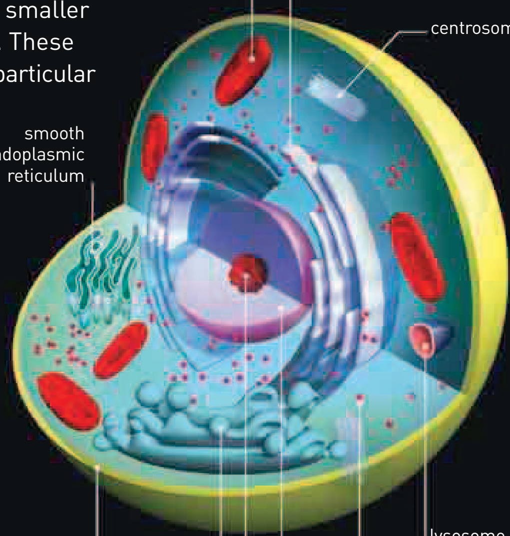
A CELL HAS COMPLEXITY ON A MINISCULE SCALE AND IS THE SMALLEST THING ALIVE

A plant or animal body contains more cells than people who have ever lived—hundreds could fit on a pinhead. Within its outer structure, a cell has chemistry of unrivaled intricacy—to manage its growth, reproduction, and nourishment.

THEODOR SCHWANN
German scientist Schwann, a founding father of Cell Theory, insisted that cells could be understood in chemical terms with no mysterious "life force."

The first cells were seen in 1663 when English scientist Robert Hooke found cork cells with his microscope. But it wasn't until the nineteenth century that their significance was better appreciated and German biologists developed a "Cell Theory." They suggested that cells were the units of all organisms and could arise only from other cells. In other words, cellular life could not form spontaneously. By 1900, scientists began to see how this reproductive ability was linked to the cell's nucleus and its chromosomes. This work would culminate with the discovery that a self-copying chemical inside the nucleus—called DNA—lay at the heart of the process.

While some cells, such as those of bacteria, are structurally simple, those of animals and plants contain even smaller compartments for specific roles. These so-called organelles enclose the particular mixture of chemicals needed for a specific job.

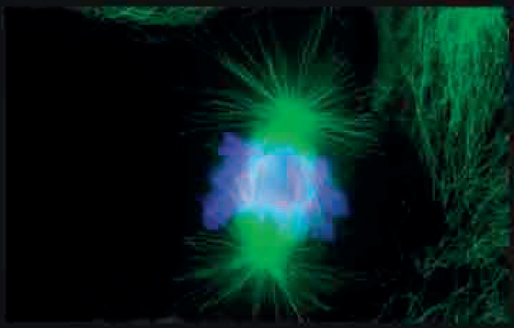


ANIMAL CELL
Most animal cells are smaller than plant cells because they lack a rigid supporting cell wall. This also makes animal cells less angular in shape. Many organelles found in plant cells (right) are also found in animal cells.

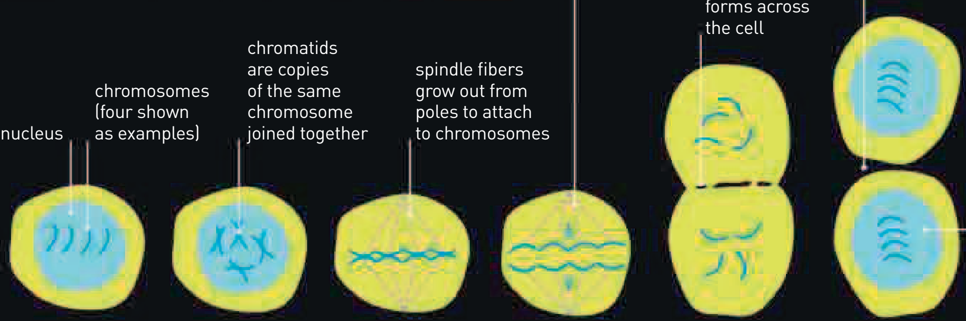
60 TRILLION THE POSSIBLE NUMBER OF **CELLS** THAT MAKE UP **THE HUMAN BODY**

CELL DIVISION

By the late 1800s, microscopes were good enough to scrutinize dividing cells. Scientists saw threadlike chromosomes moving around in precise ways—chromosomes are bundles of the DNA that carry the information for producing new cells. Just before cell division, the cell's DNA doubles up by self-copying—so when body cells divide during growth (mitosis), each daughter cell ends up with a copied set of each chromosome, or DNA bundle. During sexual reproduction, a special kind of division (meiosis) halves the chromosome number to make sperm and eggs; the normal number is restored when the egg and sperm cells combine during fertilization.



IMMUNOFLUORESCENT MICROGRAPH
Today, scientists can reveal aspects of the cell that are invisible with conventional microscopy by using fluorescently dyed antibodies that stick to specific structures. Here, the antibodies illuminate the spindle fibers (green) that pull chromosomes (blue) around during cell division.



MITOSIS
During growth, this multistage division keeps all cells genetically identical. A system of protein cables called the spindle pulls chromosomes in such a way that daughter cells end up with the same chromosome number as the parental cell.